



CS-E4960 - Software Testing and Quality Assurance D, Lecture, 3.9.2024-26.11.2024

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Theory Assignment for Statement Coverage and MC/DC

To do: Make a submission

To do: Receive a grade

Opened: Tuesday, 24 September 2024, 10:00 AM
Due: Tuesday, 1 October 2024, 10:00 AM

White-box test design allows you to take structural information about the SUT into consideration. You may have different targets regarding how much of the SUT code you want your tests to cover. In this assignment, consider how you would design test cases to achieve 100% statement coverage and modified condition/decision coverage.

Consider the Python program in the file **discount.py** and answer the following questions based on the definitions for coverage given in the lecture slides.

Instructions

- Download the Python file **discount.py** and the HTML assignment file.
- Open the HTML file in a web browser to answer the questions:

Submission Instructions

- Save your answers by clicking the "Save Answers as JSON" button in the HTML file. This will generate a JSON file with your responses.
- Submit only the JSON file, named **studentnumber_coverage.json**, replacing "studentnumber" with your actual student number.
- Each correctly answered question is worth **0.5 points**.

Make sure to save your work frequently to avoid losing progress.

When answering the questions, do not make the coverage requirements stricter than what is defined in the lecture slides. For example, if 100% path coverage would also give 100% statement coverage, but the statement coverage criteria are weaker t han the path coverage criteria, you must answer the statement coverage questions in such a way that the answer results in a weaker coverage than the path coverage criteria would require.

Tips:

- You can draw up a simple graph of the if-elif-else structure to help you analyse the program.
- You can make a decision table with the outcomes of the conditional expressions and the different values that the variables in those expressions can take.
- If things get messy and you lose your way, split things (i.e., the code) up into smaller pieces, solve the pieces, and then see if you can get to the answer by combining the pieces.
- The python code can be run – playing around with different input values could help. On the other hand, it's also possible to complete this assignment just by reading the code and giving it some thought.
- You can disregard Python details such as default parameter values and details of rounding money.

 discount.py	23 September 2024, 12:41 AM
 StatementCoverage_MCDC.html	24 September 2024, 1:10 AM

Submission status

Attempt number	This is attempt 1.
Submission status	No submissions have been made yet
Grading status	Not marked
Time remaining	Assignment is overdue by: 162 days 4 hours
Last modified	-

Previous activity

◀ Quiz on Robustness Testing

Next activity

Quiz on Path Testing ▶



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